Math 257 Definitions and Theorems

Week 6: Modules 19-21 and Graphs

Fall 2023

19 Bases and Dimension

Definition 19.1. Let V be a vector space. A sequence of vectors $(\mathbf{v}_1, \ldots, \mathbf{v}_p)$ in V is a **basis** of V if

- $V = \text{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_p)$, and
- $(\mathbf{v}_1, \ldots, \mathbf{v}_p)$ are linearly independent.

Theorem 19.1. *Every two bases in a vector space V contain the same numbers of vectors.*

Definition 19.2. The number of vectors in a basis of V is the **dimension** of V.

Theorem 19.2. *Suppose that V has dimension d. Then*

- *A sequence of d vectors in V are a basis if they span V.*
- *A sequence of d vectors in V are a basis if they are linearly independent.*

Theorem 19.3. *A basis is a **minimal** spanning set of V; that is the elements of the basis span V but you cannot delete any of these elements and still get all of V.*

20 The Four Fundamental Subspaces

Algorithm 20.1. To find a basis for $\text{Nul}(A)$:

- Find the parametric form of the solutions to $A\mathbf{x} = \mathbf{0}$.
- Express solutions $\mathbf{x}$ as a linear combination of vectors with the free variables as coefficients.
- Use these vectors as a basis of $\text{Nul}(A)$.

Definition 20.1. The **rank** of a matrix is the number of pivots it has.

Theorem 20.1. *Let A be an $m \times n$ matrix with rank r. Then* $\dim \mathrm{Nul}(A) = n - r$.

Theorem 20.2. *Let A be an $m \times n$ matrix with rank r. The pivot columns of A form a basis of* $\mathrm{Col}(A)$. *In particular,* $\dim \mathrm{Col}(A) = r$.

Theorem 20.3. *Let A, B be two row-equivalent matrices. Then* $\mathrm{Col}(A^T) = \mathrm{Col}(B^T)$.

Theorem 20.4. *Let A be $m \times n$ matrix with rank r. Then the non-zero rows in an echelon form of A form a basis of* $\mathrm{Col}(A^T)$, *and thus* $\dim(A^T) = r$.

Theorem 20.5. *Let A be an $m \times n$ matrix with rank r. Then*

- $\dim \mathrm{Col}(A) = \dim \mathrm{Col}(A^T) = r$, *and*
- $\dim \mathrm{Nul}(A) = n - r$.

Graphs

Definition 20.2. A **graph** is a set of nodes (or: vertices) that are connected through edges.

Definition 20.3. Let $\mathcal{G}$ be a graph with n nodes. The **adjacency matrix** of $\mathcal{G}$ is the $n \times n$-matrix $A = (a_{ij})$ such that

$$a_{ij} = \begin{cases} 1 & \text{if there is an edge between node } i \text{ and node } j \\ 0 & \text{otherwise .} \end{cases}$$

Definition 20.4. A **walk of length** k on a graph of is a sequence of $k+1$ vertices and k edges between two nodes (including the start and end) that may repeat. A **path** is walk in which all vertices are distinct.

Definition 20.5. A graph is **connected** if for every pair of nodes i and j there is a walk from node i to node j. A graph is **disconnected** if it is not connected.

Theorem 20.6. *Let $\mathcal{G}$ be a graph and let A be its adjacency matrix. Then the entry in the i-th row and j-th column of A^ℓ is the number of walks of length ℓ from node j to node i on $\mathcal{G}$.*

Definition 20.6. A **directed graph** is a set of vertices connected by edges, where the edges have a direction associated with them.

Definition 20.7. Let G be a directed graph with m edges and n nodes. The **adjacency matrix** of G is the $n \times n$ matrix $A = (a_{i,j})_{i,j}$ with

$$a_{i,j} = \begin{cases} 1, & \text{if there is a directed edge from node } j \text{ to node } i \\ 0, & \text{otherwise} \end{cases}$$

Definition 20.8. Let G be a directed graph with m edges and n nodes. The **edge-node incidence matrix** of G is the $m \times n$ matrix $A = (a_{i,j})_{i,j}$ with

$$a_{i,j} = \begin{cases} -1, & \text{if edge } i \text{ leaves node } j \\ +1, & \text{if edge } i \text{ enters node } j \\ 0, & \text{otherwise} \end{cases}$$

Definition 20.9. A **connected component** of an undirected graph is a part in which any two vertices are connected to each other by paths, and which is connected to no additional vertices in the rest of the graph. The connected components of a directed graph are those of its underlying undirected graph. A graph is **connected** if only has one connected component.

Theorem 20.7. *Let $\mathcal{G}$ be a directed graph and let A be its edge-node incidence matrix. Then* $\dim \mathrm{Nul}(A)$ *is equal to the number of connected components of $\mathcal{G}$.*

Definition 20.10. A **cycle** in an undirected graph is a path in which all edges are distinct and the only repeated vertices are the first and last vertices. By cycles of a directed graph we mean those of its underlying undirected graph.

Definition 20.11. The span of all cycle vectors of a graph $\mathcal{G}$ is called the **cycle space**.

Theorem 20.8. *Let $\mathcal{G}$ be a directed graph and let A be its edge-node incidence matrix. Then the cycle space of $\mathcal{G}$ is equal to* $\mathrm{Nul}(A^T)$.

21 Orthogonal Complements

Definition 21.1. Let W be a subspace of $\mathbb{R}^n$. The **orthogonal complement** of W is the subspace $W^\perp$ of all vectors that are orthogonal to W; that is

$$W^\perp := \{\mathbf{v} \in \mathbb{R}^n \ : \ \mathbf{v} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\}.$$

Remark. Observe that $(W^\perp)^\perp = W$.

Theorem 21.1. *Let A be an $m \times n$ matrix. Then* $\mathrm{Nul}(A)$ *is the orthogonal complement of* $\mathrm{Col}(A^T)$*; that is* $\mathrm{Nul}(A) = \mathrm{Col}(A^T)^\perp$.

Remark. It follows that

- $\mathrm{Nul}(A)^\perp = \mathrm{Col}(A^T)$.
- $\mathrm{Nul}(A^T) = \mathrm{Col}(A)^\perp$.

Theorem 21.2. *Let V be a subspace of $\mathbb{R}^n$. Then* $\dim V + \dim V^\perp = n$.